

# MAJETI LAHARI

Software Developer – Generative AI & Systems

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## CAREER OBJECTIVE

As a Software Engineer specializing in **Generative AI**, **Backend Systems**, and **Cloud Engineering**, I design and deploy scalable AI-driven production grade systems that address real-world challenges. With a passion for **Generative & Agentic AI**, and cloud solutions, I seek opportunities to build reliable, impactful AI systems in enterprise.

## TECHNICAL SKILLS

### Generative AI & LLMs:

- Large Language Models, Retrieval Augmented Generation (RAG), LangChain, Agentic AI Systems
- Prompt Engineering & Templates, Vector Databases, Embeddings, Chunking, and Metadata Design

### AI & ML Foundations:

- Fundamentals of Machine Learning, Natural Language Processing (NLP), Deep Learning

### Backend & Cloud Engineering:

- Python, Flask, IBM Cloud, OpenShift, Docker, Jenkins, Tekton
- IBM Cloud Object Storage (COS), DevOps Automation & Monitoring (Shell Scripting)

### Security & Reliability:

- Secrets Management (IBM Cloud Secrets Manager), Security Scanning (Twistlock, Mend)
- Branch Protection & Access Control (GitHub)

### Developer Tools & Workflow:

- GitHub, Postman, VS Code, Jupyter Notebooks, Agile Development, CI/CD Integration

## PROFESSIONAL EXPERIENCE

IBM – Software Developer & AI/DevOps Engineer

January 2023 – Present

**Previous Role:** Software Developer | **Current Role:** DevOps Engineer (April 2025 - Present)

### AI Development (Throughout Tenure):

- Project Modules Q&A Chatbot:**  
Built a **RAG-based chatbot** that enables developers and SREs to query **GitHub-hosted project documentation** using natural language. Integrated **vector search**, **CI/CD**, and **cloud deployment** for seamless performance.
- Developer's Genie (Agentic AI System):**  
Developed an **agentic GenAI system** for **conversational understanding**, **summarization**, and **comparison** of evolving **technical specifications** and **codebases** across multiple versions.
- Run Status Bot:**  
Created a **natural-language chatbot** that queries **IBM Db2** to retrieve **structured run-status data**, using **LLMs** to generate human-readable summaries for operational teams, enhancing system health monitoring.

- **SRE Support Bot:**  
Built a **semantic-search-enabled** chatbot that ingests **runbooks** and **operational procedures**, enabling **SREs** to query troubleshooting knowledge conversationally, accelerating incident triage.
- **LLM-Based Test Case Generator:**  
Implemented a **GenAI system** that auto-generates **unit** and **integration tests** directly from **feature specifications** and **code logic**, improving test coverage and automation.
- **Incident Management & Resolution Intelligence System:**  
Developed an **AI-powered system** that leverages **historical incident** and **resolution data** to suggest probable **root causes** and **resolution actions** during live operational incidents.

#### Cloud, DevOps & AI Infrastructure (April 2025 - Present):

- **CI/CD Pipelines:**  
Maintained and optimized **CI/CD pipelines** using **Jenkins** and **Tekton** for **automated build, test, and deployment** of applications and AI systems, ensuring smooth and reliable updates.
- **Security Integration:**  
Integrated **security scanning** tools like **Twistlock** and **Mend** into pipelines to enforce **container** and **dependency compliance**, maintaining secure and production-grade environments.
- **GitHub Management:**  
Managed **GitHub access controls, branch protection, secrets management, and vulnerability workflows** to ensure secure and efficient code development across teams.
- **Automation & Scripting:**  
Automated key workflows via **shell scripting** for tasks such as **file operations, DB backups, monitoring alerts, and cron job scheduling**, improving operational efficiency.
- **Cloud Deployments:**  
Deployed applications on **IBM Cloud**, including setting up **IAM policies**, managing **Object Storage, OpenShift, Secrets Manager**, and integrating with **LogDNA** for enhanced logging and monitoring.

#### Software Development (January 2023 - April 2025):

- **Full-Stack Web Applications:**  
Developed **full-stack web applications** with dynamic **UIs** using **Angular** and robust backend APIs with **PHP Laravel**, providing interactive user experiences.
- **Database Management:**  
Managed **IBM Db2** database operations and schema design, implementing optimized **SQL queries** for high-performance and data integrity across systems.
- **Data Consistency & Integrity:**  
Ensured **data consistency** and **integrity** across distributed systems, working with cross-functional teams to maintain high data reliability.
- **Automated UI Testing & QA:**  
Automated **UI testing** and **quality assurance** processes, integrated within **CI/CD pipelines** for **regression testing** and coverage across releases.
- **Agile Development:**  
Actively collaborated in **Agile workflows**, including sprint planning, daily stand-ups, and code reviews with cross-functional teams, fostering an adaptive and efficient work environment.

## PROJECTS

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### MCQ Generator (Multi-Modal GenAI)

**Tech Stack:** Python, Streamlit, Mistral LLM, OCR, Whisper, MoviePy

A GenAI system that converts text, documents, images, audio, and video into high-quality MCQs with answers and explanations. Designed with modular pipelines, input validation, and educational content filtering.

### Developer's Genie – Agentic AI Assistant

**Tech Stack:** Watsonx.ai, LangChain Agents, React, SQLite, Multi-Agent Design

An award-winning agentic AI system that enables conversational understanding of multiple versions of technical documents and code. Supports summarization, comparison, debugging assistance, and serves as a KT tool for teams.

### Q&A Chatbot on Project Modules (RAG)

**Tech Stack:** Watsonx.ai, Milvus, Flask, IBM Cloud, Docker, Jenkins, COS

An end-to-end Retrieval-Augmented Generation application that allows developers and SREs to query project documentation conversationally. Includes automated ingestion, embeddings, CI/CD, containerized deployment, and cloud hosting.

### Portfolio Chatbot:

**Tech Stack:** React, Flask, LangChain, Llama 3.1, ChromaDB, Sentence Transformers, RAG

My digital companion, an AI-powered chatbot integrated into my portfolio that provides instant answers about my projects, skills, and professional experience using retrieval-augmented generation (RAG) and local LLM inference

### Enterprise AI Utilities

**Tech Stack:** Watsonx.ai, IBM Db2, Prompt Engineering, NLP

Built internal AI tools including a Run Status Bot for real-time system introspection via Db2 and an LLM-based test case generator to improve coverage by auto-generating unit and integration tests from specs.

## EARLY WORK

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Machine Learning-Based Housing Price Prediction System.  
Named Entity Recognition (Bi-LSTM).  
Certified Machine Learning intern | Pantechelearning.  
Room Allotment System: Python + DB integration for automated allocations.  
Static Web Page (SVLH Fashions): Responsive web page for client business.

## EDUCATION

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|------------------------------------------------------------------------------------------------------------------------|------------------------|
| Ramachandra College of Engineering, Eluru   JNTU-Kakinada<br>Bachelor of technology - Computer Science and Engineering | 2019-2023   CGPA: 8.14 |
| Intermediate, Sri Chaitanya Junior College, Eluru   AP Board<br>MPC                                                    | 2017-2019   GPA: 9.94  |
| Secondary Education, Little Buds High School, Eluru   AP Board                                                         | 2016-2017   GPA: 10    |

## STRENGTHS

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Passionate, Adaptive, Quick Learner, Mentorship, Collaborative, Tech Enthusiast